



Social Transformation Fellowship: Skin of Color Catered Machine Learning Model

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Abstract

- We found a problematic discrepancy in the diagnosis Skin of Color (SoC) and non-SoC in dermatological settings, which was a need that other projects were not focusing on.
- Our team is constructing a ML-powered diagnostic tool that can classify skin conditions frequently misdiagnosed on SoC, currently starting with eczema.
- Our model surpassed an industry competitor on relevant performance metrics for image classification by dataset augmentation.
- The range we used to classify skin tones is known at the **Fitzpatrick scale** with tones ranging from F1-F7 with F1 being the fairest and F7 being the deepest.

Problem and Context

- ML-powered dermatology tools to reduce diagnostic errors and improve access to care. Current tools often replicate existing disparities in healthcare, especially when diagnosing individuals with Skin of Color (SOC).
- Dermatological disparities in the U.S. are caused by the underrepresentation of SoC in medical training resources. According to Narla et al. (2022), "47% of dermatologists reported feeling inadequately trained to diagnose conditions on skin of color".

Stakeholder Research

- Our team interviewed medical students, a nursing student and healthcare professionals in order to gain insight on:
 - Disparities in SoC Healthcare education
 - Sentiments surrounding artificial intelligence in healthcare
- Our stakeholders said:

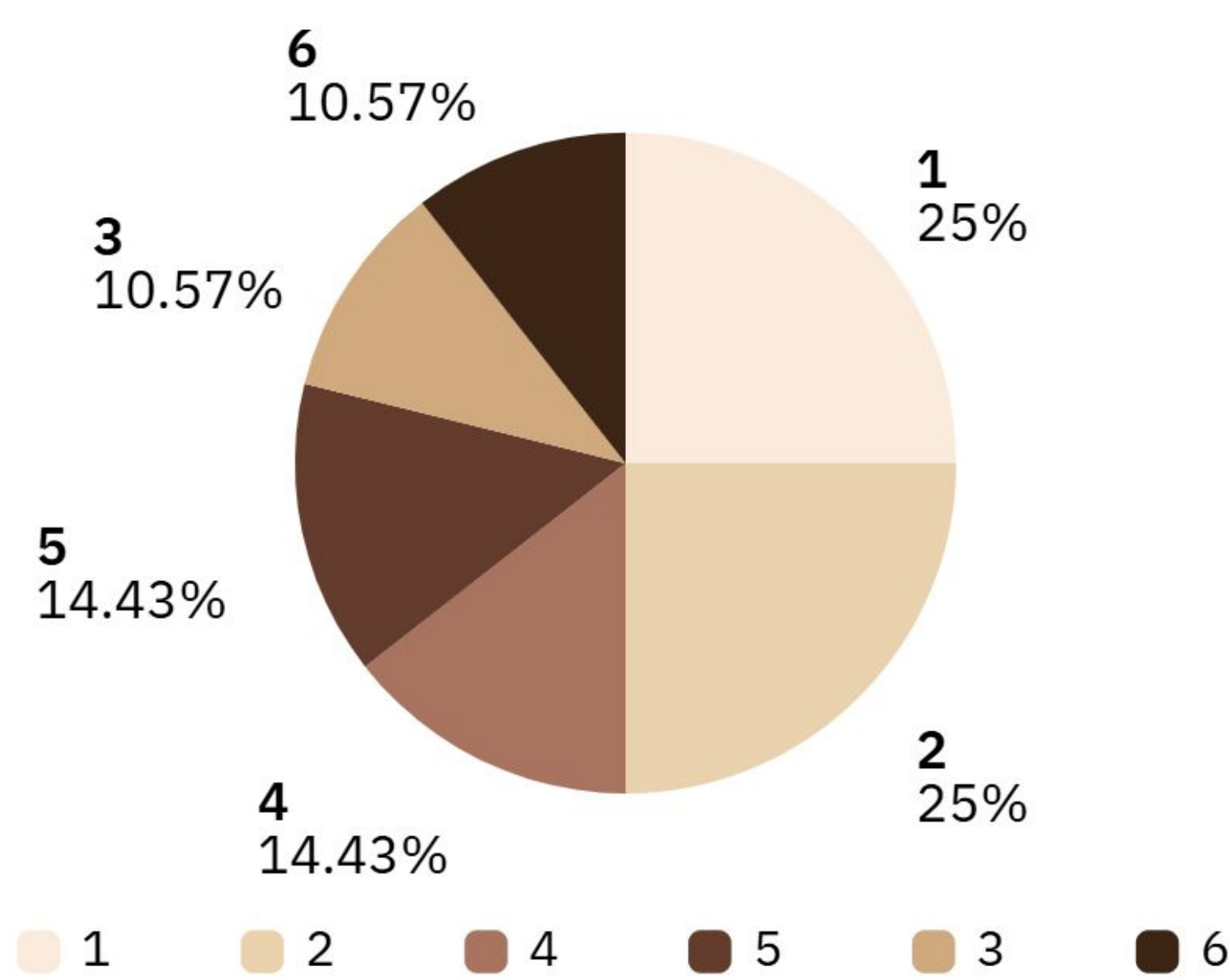
“ML classification tools would be useful in healthcare spaces that don’t have specialists”

“... not much in-depth content on what wounds and other conditions look like on darker skin was available”

Our Implementation

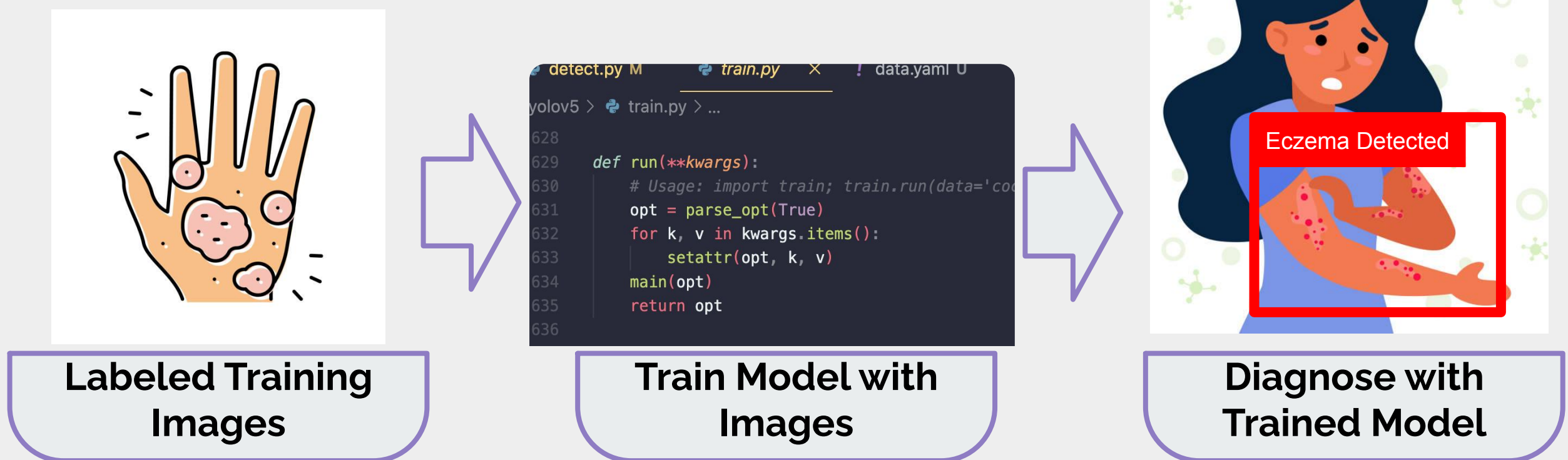
Step 1: Assembling a Representative Dataset

Distribution of skin tones across SoC Dataset in terms of Fitzpatrick Scale Rating



Our team ensured that our training and testing dataset consisted of 50% images of SoC which is represented by the Fitzpatrick scale ratings 3-7 for both images featuring eczema as well as images containing other similar conditions such as bacterial skin infections.

Step 2: Training our Model



Then we trained a standard ResNet model on the augmented dataset. In this process our model imitates a human brain, rapidly learning underlying patterns that are key distinguishing between eczema and other conditions.

Step 3: Evaluate against competitor

Finally, we evaluated the model against a testing set that featured images on a range of Fitzpatrick skin types **F1-F7**. We then compared our model's performance to another commonly used dermatological model built on ResNet.

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References

1. Narla, S., Heath, C. R., Alexis, A., & Silverberg, J. I. (2023). Racial disparities in dermatology. *Archives of dermatological research*, 315(5), 1215–1223. <https://doi.org/10.1007/s00403-022-02507-z>
2. Adelekun, A., Onyekaba, G., & Lipoff, J. B. (2021). Skin color in dermatology textbooks: An updated evaluation and analysis. *Journal of the American Academy of Dermatology*, 84(1), 194–196. <https://doi.org/10.1016/j.jaad.2020.04.084>

Results

Performance of SoC catered Training Model vs Standard Dermatological model

Metric	SoC-catered model			Standard Dermatological Classification model		
	All Skin (F1-F7)	SoC skin (F3-F7)	non-SoC skin (F1-F2)	All Skin (F1-F7)	SoC skin (F3-F7)	non-SoC skin (F1-F2)
Accuracy	0.83	0.85	0.82	0.71	0.67	0.74
Precision	0.83	0.77	0.85	0.96	0.29	0.99
F1 Score	0.90	0.77	0.86	0.61	0.04	0.75
Auroc Score	0.90	0.92	0.88	0.72	0.50	0.79

Relevance of Metrics in Context of Classifying Eczema

Metric	Meaning
Accuracy	Correct predictions / total predictions
Precision	Correct eczema predictions / all eczema predicted; our model shows increased precision
F1 Score	How well model balances missing vs. false alarms
AUROC Score	Ability to distinguish eczema vs. non-eczema

Discussion

- **Outcome** - Our representative dataset augmentation, as dictated by the Fitzpatrick scale, led to higher performance in classification of eczema on SoC.
- **Problems** - Our biggest hurdle was finding substantial, representative datasets, especially in classification of other, rare conditions. Getting clearance for data from hospitals also proved to be difficult and slow. We've developed our challenge resilience habits, which helped us to move forward.
- **Next Steps** - We'll experiment with dataset augmentation and other bias-reduction techniques to further optimize the model's performance. We plan to develop a web application for providing reliable diagnosis for multiple conditions.